Jay Hemnani

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Gujarat, India (Open to Relocate)

Education

Pandit Deendayal Energy University (PDEU) 2018 – 2022 | Gandhinagar, India
B.Tech in Computer Engineering | GPA: 8.7/10

- **Coursework:** Database Systems, Big Data Analytics, Operating Systems, Software Development.

Technical Skills

Programming: Python, SQL, Java, Go, JavaScript/TypeScript
Data Engineering: Apache Kafka, Apache Airflow, PySpark, Pandas, ETL/ELT Pipelines, dbt, Data Modeling
Databases: PostgreSQL, MySQL, MongoDB, TimescaleDB, Redis, Data Warehousing
Cloud & Infrastructure: AWS (S3, EC2, Lambda, Glue), GCP (BigQuery), Docker, Kubernetes
Streaming & Processing: Kafka Streams, Spark Streaming, Batch Processing, Event-Driven Architecture
Tools: Tableau, Power BI, Streamlit, Dash, Git, GitHub Actions (CI/CD), Prometheus

Experience

Data Analyst, *Elite Hotel Group* Summer 2025 | Remote

- Defined **metrics and SLAs** with finance and operations; they became the spec for the ETL and forecasting work.
- Built **automated ETL pipelines** in SQL and Python, cutting manual data preparation by **40%** and making multi-property analytics more consistent.
- Built **dashboards** in Tableau and Power BI for occupancy, revenue, and KPI tracking, and **time-series demand forecasting models** for dynamic pricing.

Technical Consultant, *Independent* 2022 – 2024 | Remote

- Did **data analytics and pipeline consulting** for small businesses; built their reporting automation and data infrastructure.

AI/ML Intern, *Amnex* Jan – May 2022 | Gujarat, India

- Built a **fraud detection ensemble** (Random Forest, XGBoost, SMOTE for class imbalance); **94% precision** on transaction data.
- Built **automated analytics dashboards** on statistical models for operational reporting and anomaly detection.

Software Engineering Intern, *Cactus Creatives* May – Nov 2019 | Gujarat, India

- Built a **cloud-native communication platform** for first responders (microservices on Azure); set up **CI/CD pipelines** that cut deployment time by **60%**.

Projects

Stock Data Platform | *Python, Kafka, Airflow, TimescaleDB, Chart.js, Docker*

- A **Kafka producer** polls live prices every 15s in market hours; the consumer upserts into TimescaleDB in batches (50 rows or 10s) and reconnects with backoff on failure.
- **23 Airflow DAGs** run batch ETL, backfills and aggregations over 4 sources (yfinance, SEC EDGAR, FRED, Alpaca). Loads are incremental from the last loaded date, so reruns are idempotent.
- **10-table star schema** (3 dimension, 7 fact) holding full price history for 10 tickers plus fundamentals, earnings, filings and macro data. Chart.js dashboard, 7 Docker services.

Kayak Travel Metasearch (team) | *Node.js, React, FastAPI, MySQL, MongoDB, Redis, Kafka, Docker*

- **3-tier system** with an API gateway (JWT auth, rate limiting) routing to user, search and booking services.
- Data split by workload: **MySQL** for transactional bookings, **MongoDB** for analytics and logs, **Redis** for search caching.
- Put **Kafka** behind an interface with an in-memory stand-in, so the team could build against topics before a broker existed.

Website Watcher | *Python, Flask, ArchiveBox, SQLite FTS5, APScheduler, Docker*

- Crawls sites on a schedule (every 2h, robots-compliant), snapshots pages with ArchiveBox, detects changes by **SHA-256 hash**, and makes every version full-text searchable.
- Runs on Docker Compose or systemd with health checks and **Prometheus metrics**; Merkle trees and OpenTimestamps prove when a version existed. 26 Python modules.

Open Source Contributions

270+ merged pull requests across 51 open-source projects, including vLLM, Effect, Caddy, Lightpanda, radare2, SeaweedFS, Jaeger, Home Assistant and NestJS.
A plain-English note on each one: jayhemnani.in/oss